

Nadjiba Rahal

Backend Developer | C / ASP.NET Core | AI & Intelligent Systems

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Profile

Computer Science Engineer specializing in **Intelligent Systems and Data Science**, experienced in building end-to-end **backend, AI and data systems**. Hands-on experience with **C#, .NET, ASP.NET Core, REST APIs, PostgreSQL and Docker**, alongside Machine Learning, Computer Vision, NLP and AI agents. Industrial experience at **Sonatrach and Sonelgaz**, with research in Neural Architecture Search and Medical Vision-Language Models.

Technical Skills

Backend: C#, .NET 10, ASP.NET Core, EF Core, REST APIs, FastAPI, Django, Node.js, Express, Git
Data: SQL, PostgreSQL, MongoDB, Oracle, DuckDB, Python, Pandas, NumPy, ETL, Power BI
AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG, AI Agents, VLMs, embeddings, semantic search, OCR, PyTorch
Cloud / DevOps: Docker, Azure Container Apps, GitHub Actions, CI/CD, Linux, Bash, TCP/IP

Professional Experience

Data Analyst Intern Sep. – Dec. 2025
Sonatrach Exploration Division, Boumerdès

- Developed **Power BI dashboards** for operational analysis; used **SQL and Python** to clean data and define KPIs.

IT / Systems Intern Aug. 2025
Sonelgaz Cap Djinet Power Plant, Boumerdès

- Supported IT activities, including systems monitoring, maintenance and information management.

Selected Projects

ISHARA Algerian Sign Language AI System 2026
C# .NET 10 ASP.NET Core EF Core PostgreSQL ONNX Next.js

- Built an end-to-end AI platform with a **REST backend**, PostgreSQL data layer, video processing, MediaPipe and multi-agent generation pipeline.

Bellevue Medical Clinic Management Platform 2026
Next.js TypeScript Digital Patient Record Medical UI

- Designed a practice platform centered on **Digital Patient Records**, with workflows for consultations and appointments.

Agentic Video Assistant 2026
Python smolagents Gradio Docker DuckDuckGo

- Built an LLM agent that researches ideas, enhances prompts, and orchestrates asynchronous video generation.

Enterprise RAG Engine & AI Job Intelligence Agent 2026
Python FastAPI PostgreSQL ChromaDB OCR Embeddings

- Built modular AI backends combining **OCR, semantic search, retrieval and multilingual processing**.

Hardware-Aware Neural Architecture Search 2025 – 2026
PyTorch Multi-objective Optimization CNN

- Designed NAS optimizing **accuracy and latency**, evaluating 15,000 CNN architectures and analyzing the Pareto front.

SaaS E-Commerce Platform 2026
Next.js TypeScript MongoDB NextAuth Cloudinary

- Built a full-stack e-commerce platform with authentication, media management, and delivery across 58 wilayas.

Education

State Engineer's Degree in Computer Science Intelligent Systems & Data Science 2024 – 2027
École Nationale Supérieure d'Informatique (ESI)

Preparatory Computer Science Program 2022 – 2024
ESTIN, Béjaïa

Certifications & Languages

Certifications: Azure Container Apps (2026) | Hugging Face Agents (2026) | EF SET C1 English (2026) | Python for Everybody (2025)
Languages: Arabic (Native) | French (Professional) | English (C1) | German (A2)